Relation Between Fermi–Amaldi and Exact Exchange Energy Densities

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I. MAIN TEXT

Consider the exact exchange energy density (closed-shell):

$$e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) = -\frac{1}{4}\int \frac{|\gamma(\mathbf{r};\mathbf{r}')|^2}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}', \tag{1}$$

where

$$\gamma(\mathbf{r},\mathbf{r}') = 2\sum_{i=1}^{N/2}\varphi_i^*(\mathbf{r})\varphi_i(\mathbf{r}') \tag{2}$$

is the one-electron reduced density matrix [1, 2], and the Fermi–Amaldi [3, 4] energy density expressed via electron density, $\rho(\mathbf{r}) \geq 0$, and electron number, N,

$$e_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}) = -\frac{1}{2N}\int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}'. \tag{3}$$

For one-orbital systems (one-electron and two-electron singlet), the two quantities are equal for all values of $\mathbf{r}$ [5–7]:

$$e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) = e_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}) \quad N \leq 2 \quad \text{for all } \mathbf{r}. \tag{4}$$

For systems with more than one molecular orbital, the spatial behavior of both quantities needs to be studied closer to establish a relationship between the exact and the Fermi–Amaldi exchange energy densities.

For two integrals with different integrands $f(x)$ and $g(x)$, on the interval $[a, b]$

$$\int_a^b f(x)dx \geq \int_a^b g(x)dx, \quad \text{when } f(x) \geq g(x). \tag{5}$$

Therefore, let

$$e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) = \frac{1}{2}\int f(\mathbf{r},\mathbf{r}')g(\mathbf{r},\mathbf{r}')d\mathbf{r}', \tag{6}$$

$$e_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}) = \frac{1}{2}\int \tilde{f}(\mathbf{r},\mathbf{r}')g(\mathbf{r},\mathbf{r}')d\mathbf{r}', \tag{7}$$

where $f(\mathbf{r},\mathbf{r}') = -|\gamma(\mathbf{r},\mathbf{r}')|^2/2$, $\tilde{f}(\mathbf{r},\mathbf{r}') = -\rho(\mathbf{r})\rho(\mathbf{r}')/N$, and $g(\mathbf{r},\mathbf{r}') = |\mathbf{r}-\mathbf{r}'|^{-1}$. Since $g(\mathbf{r},\mathbf{r}') > 0$ everywhere in space and is the same for both exchange energy densities, to establish a relation between $e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r})$ and $e_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r})$, it would suffice to establish a relation between $f(\mathbf{r},\mathbf{r}')$ and $\tilde{f}(\mathbf{r},\mathbf{r}')$.

By the Cauchy–Schwarz inequality:

$$\rho(\mathbf{r})\rho(\mathbf{r}') \geq |\gamma(\mathbf{r},\mathbf{r}')|^2 \quad N > 2 \quad \text{for all } \mathbf{r},\mathbf{r}'. \tag{8}$$

However, the Fermi–Amaldi global prefactor, $1/(2N)$, is strictly smaller than the exact exchange prefactor, $1/4$, when $N > 2$. Because these opposing conditions compete, neither energy density integrand strictly dominates the other everywhere in space.

As $|\mathbf{r}| \to \infty$, the denominator can be approximated as $|\mathbf{r}-\mathbf{r}'| \approx |\mathbf{r}|$, revealing that both exchange energy densities approach the same asymptotic limit:

$$e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) \sim e_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}) \sim -\frac{\rho(\mathbf{r})}{2|\mathbf{r}|}. \tag{9}$$

Thus, in the asymptotic limit:

$$\lim_{|\mathbf{r}|\to\infty}\frac{e_{\mathrm{x}}^{\mathrm{FA}}}{e_{\mathrm{x}}^{\mathrm{exact}}} = 1. \tag{10}$$

Conversely, in the limit of infinitesimally small inter-electronic distances ($\mathbf{r} \to \mathbf{r}'$), the spatial parts of both integrands approach equality:

$$|\gamma(\mathbf{r},\mathbf{r}')|^2 \to \rho(\mathbf{r})\rho(\mathbf{r}') \quad \text{as} \quad \mathbf{r}' \to \mathbf{r}. \tag{11}$$

However, because the exact exchange prefactor, $1/4$, is larger than the Fermi–Amaldi prefactor, $1/(2N)$, for $N > 2$, the region where $|\mathbf{r}-\mathbf{r}'| \to 0$ is dominated by the exact exchange energy density:

$$\lim_{|\mathbf{r}|\to|\mathbf{r}'|}\frac{e_{\mathrm{x}}^{\mathrm{FA}}}{e_{\mathrm{x}}^{\mathrm{exact}}} = 0 \quad \text{for large } N. \tag{12}$$

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